

Load Growth and Distributed PV Hosting Capacity Study for a 20kV Distribution Feeder

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1 Introduction

Distribution feeders are experiencing increasing load demand while distributed PV systems are becoming more common. Unlike traditional operation where power flows from the grid to consumers, distributed PV can reduce grid demand, and in high generation conditions, export power back into the network. Voltage regulation and thermal loading are important operational constraints in feeder operation. **This project investigates the effect of increasing load demand and distributed PV integration on feeder voltage, thermal loading, losses and operating feasibility using a simplified 20 kV load-flow model in Python (pandapower package).**

2 System Model

The system under consideration consists of a slack bus connected to a PQ bus via a 2.5 km, 20 kV feeder (NAYY 4x50 SE). Bus 2 acts as the PQ load bus. Additionally, a distributed PV is placed at bus 2. The reactive demand of the load is 30% of the real power demand. The distributed PV is modelled using static generation and analysis is done using the Newton-Raphson load flow method.

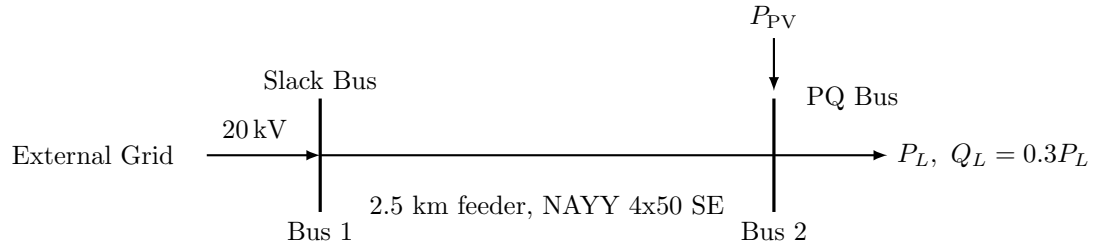


Figure 1: System Model

Table 1: Simulation Parameters

Parameter	Value
Voltage level	20 kV
Feeder length	2.5 km
Cable type	NAYY 4x50 SE
Load range	0–12 MW
Reactive demand	$Q = 0.3P$
PV range	0–11 MW
Solver	Newton–Raphson

3 Method

A load sweep is performed by varying the real load power P_L and monitoring the bus 2 voltage, line loading and line losses. This is done for two cases: no PV and 1 MW PV at bus 2. Next, both P_L and the PV power were varied to generate heatmaps for bus 2 voltage and line loading. Lastly, a binary feasible operating region heatmap was generated for the following constraint definitions:

$$0.95 \leq V \leq 1.05 \quad (1)$$

$$\text{Line Loading} = \frac{I}{I_{\max}} \leq 100\% \quad (2)$$

4 Results

4.1 Load Sweep: Voltage Response

When the load power is increased at bus 2, we demand more current. This creates a larger voltage drop across the feeder line which leads to a lower bus 2 voltage. Adding the distributed PV reduces voltage drop as shown in the diagram.

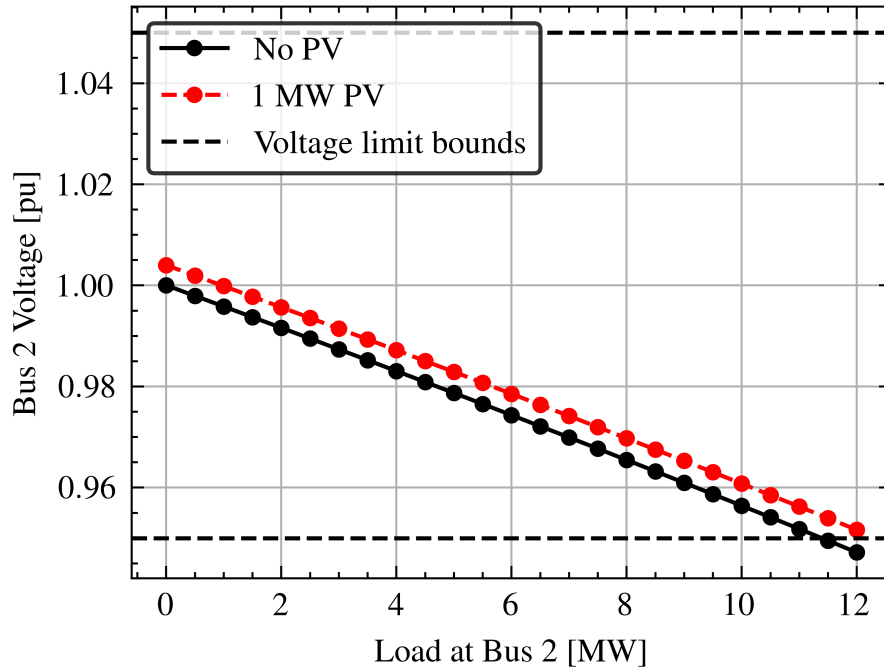


Figure 2: Bus 2 voltage response as a function of load demand for cases with and without distributed PV.

4.2 Load Sweep: Line Loading

The line loading increases linearly with load. This is because in 3-phase systems,

$$I \approx \frac{P}{\sqrt{3}V \cos \phi}. \quad (3)$$

Here, the voltage at bus 2 varies between 1 and 0.95 in the simulation range so is approximately constant. The power factor is constant as well since the simulation takes $Q = 0.3P$:

$$\text{pf} = \frac{P}{\sqrt{P^2 + Q^2}} = \frac{P}{\sqrt{P^2 + (0.3P)^2}} = \text{constant}. \quad (4)$$

Therefore, the denominator of Equation 3 is approximately constant and so $I \propto P$. The PV plot shows an initial non-zero line loading due to the PV supplying excess power and it flowing back into the grid. This drops as the load demands more power and reaches the minimum line loading at a 1 MW demand since that is the static supply of the PV. However, the line loading at this point is non-zero which is due to the reactive power requirement of $Q_L = 0.3P_L$, so the grid must still supply this small amount. As the load exceeds 1 MW, the line loading increases since the PV is at its maximum usage capacity and more current must be sourced from the slack bus.

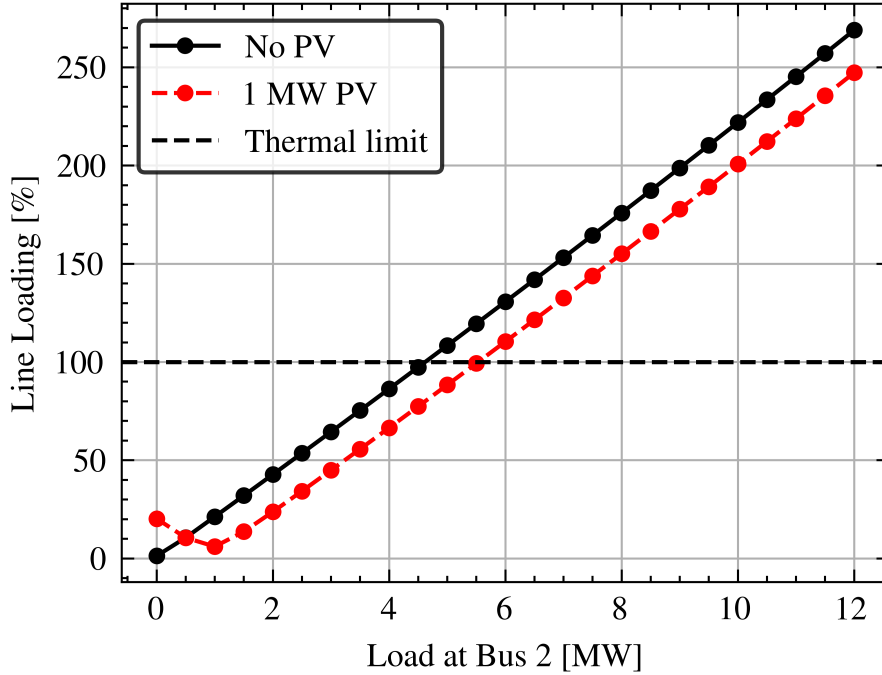


Figure 3: Feeder line loading as a function of load demand for cases with and without distributed PV.

The first thermal violation occurred at approximately 4.7 MW without PV and approximately 5.6 MW with 1MW of distributed PV. This corresponds to an increase of approximately 0.9MW in allowable load before thermal overload. The value differs slightly from 1 MW due to the finite 0.1 MW resolution of the parameter sweep.

4.3 Load Sweep: Line Loss

The feeder losses are caused by resistive heating and are approximately given by

$$P_{\text{loss}} = 3I^2R \quad (5)$$

where I is the phase current and R is the line resistance per phase. For a balanced three-phase system,

$$I = \frac{P_L}{\sqrt{3}V \cos \phi}, \quad (6)$$

and substituting this into the loss equation gives

$$P_{\text{loss}} = 3I^2R \quad (7)$$

$$= 3 \left(\frac{P_L}{\sqrt{3}V \cos \phi} \right)^2 R \quad (8)$$

$$= \frac{P_L^2 R}{V^2 \cos^2 \phi} \quad (9)$$

$$\propto P_L^2 \quad (10)$$

assuming approximately constant voltage and power factor over the operating range. This explains the approximately quadratic behaviour observed in the graph. Adding distributed PV reduces the net power imported through the feeder, decreasing current flow and therefore reducing line losses.

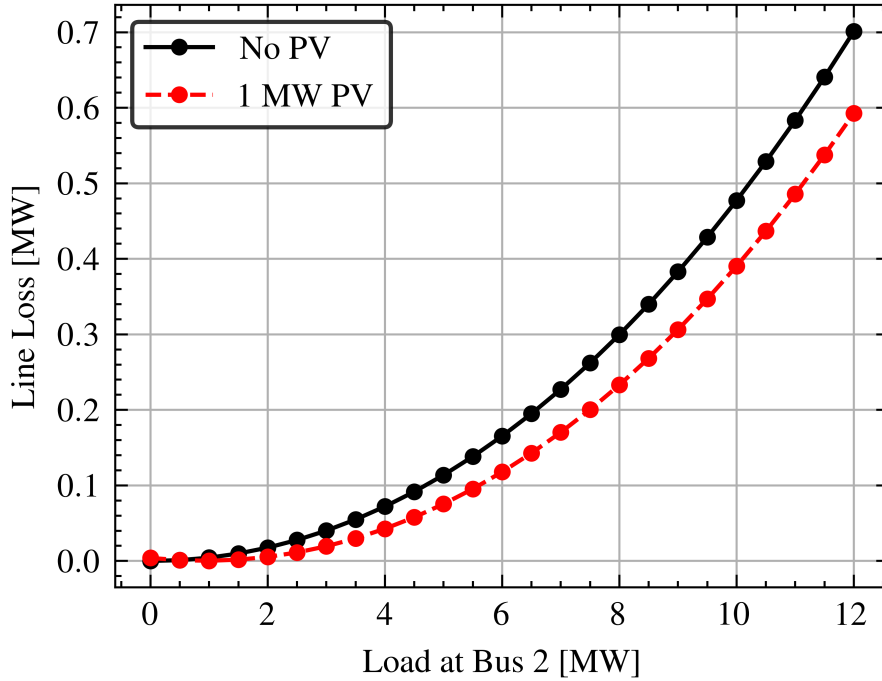


Figure 4: Real power losses in the feeder line as a function of load demand.

4.4 Load-PV Voltage Heatmap

A heatmap was generated by sweeping the PV and Load from 0-11 MW to monitor the bus 2 voltage. Under this model, higher PV increases bus voltage and higher load decreases bus voltage. At zero load and PV at 6 MW, the bus 2 voltage increases past 1 pu indicating reverse power flow from the feeder toward the external grid. For these simulation parameters, no constraints were violated with voltage.

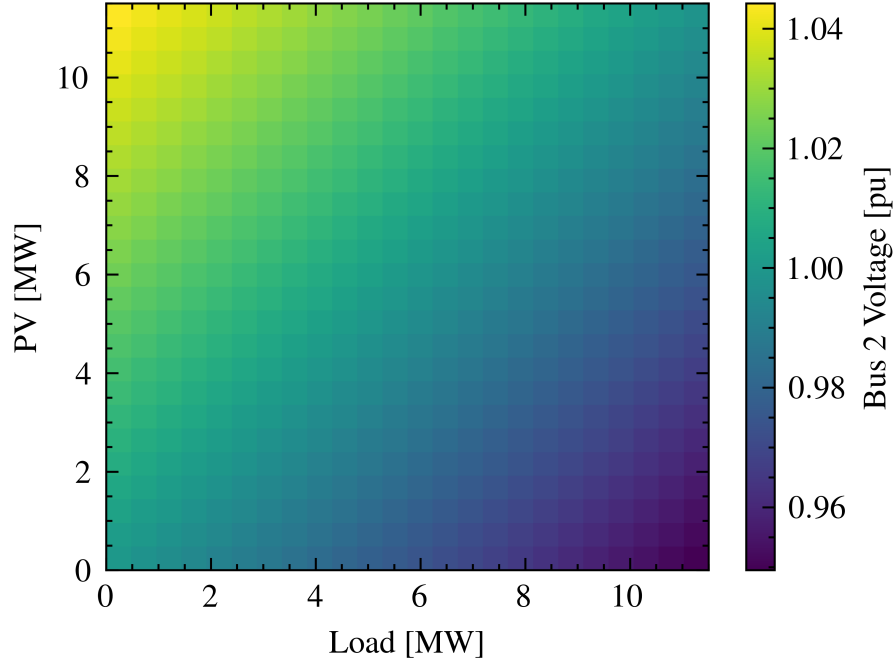


Figure 5: Voltage heatmap showing Bus 2 voltage magnitude across different load and PV operating points.

4.5 Load-PV Loading Heatmap

The same heatmap was generated but this time with the line loading as the monitored parameter. The line loading is approximately governed by the net power transfer between Bus 2 and the external grid, this gives three broad operating behaviours:

- **High PV/low load:** PV overgenerates so the line loading is high due to power flow towards grid.
- **Low PV/high load:** PV under generates so the grid supplies most of the power, line loading is due to power flowing from the grid.
- **High PV $\approx$ low load:** The power from the PV adequately supports the load, reducing line loading since insignificant power flows from the grid.

The line loading condition is violated for values of PV and Load that are far apart in value, shown by the dark lines.

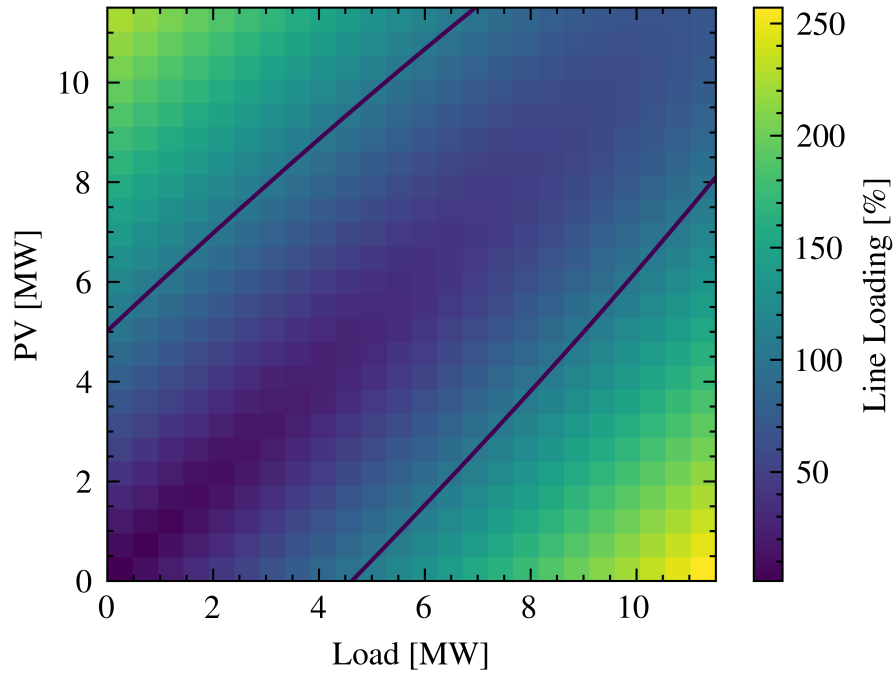


Figure 6: Feeder loading heatmap across varying load and distributed PV operating conditions.

4.6 Feasible Operating Region

This heatmap was created to show which operating regions were allowed under the voltage and line loading constraints. Line loading is the limiting constraint for the system being considered since the regions match up with the constraint lines for line loading in the previous section.

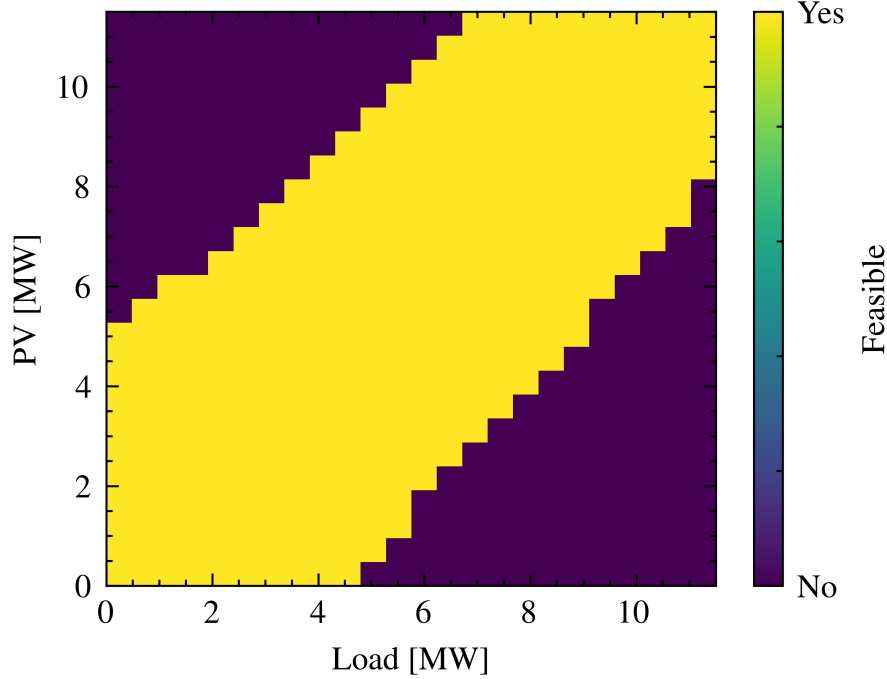


Figure 7: Feasible operating region considering voltage and thermal loading constraints.

5 Limitations

This study used a simplified two-bus balanced feeder and therefore does not capture the complexities of multiple buses and branches in real networks. Simulations were steady-state only and did not include time-varying load or PV output. Future work could extend the study to multi-bus feeders and time-series analysis.

6 Conclusion

This study investigated the impact of load growth and distributed PV integration on a 20 kV distribution feeder using NR power-flow analysis. Results showed that distributed PV reduced feeder loading and losses, and increased allowable load before thermal overload. The first thermal violation shifted from approximately 4.7 MW to 5.6 MW with 1 MW PV as expected. Voltage constraints were not violated within the operating range considered, which means thermal loading was the dominant constraint.